

L8: Dynamic Communities

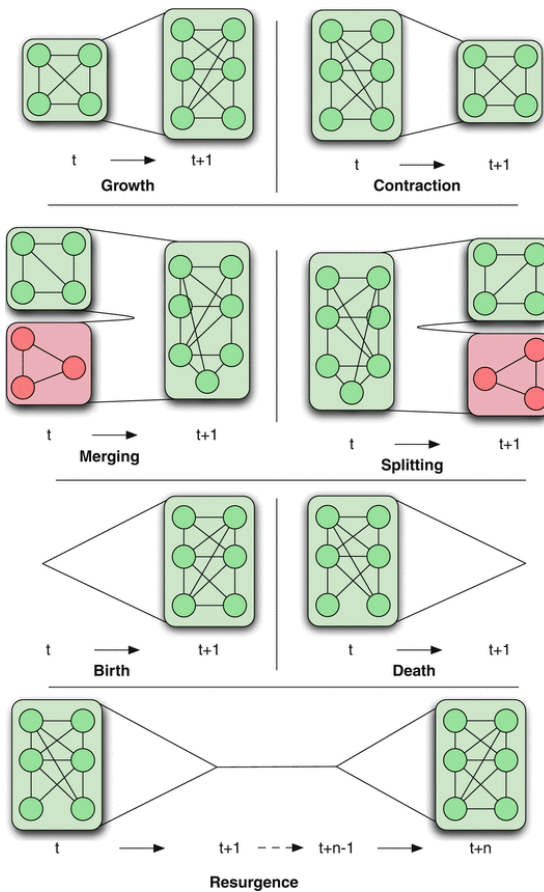


Figure 1 [Dynamic Community Detection](https://doi.org/10.1007/978-1-4614-6170-8_383) https://doi.org/10.1007/978-1-4614-6170-8_383 by Cazabet R., Amblard F.

In practice, many networks change over time through the creation or removal of nodes and edges. This dynamic behavior also causes changes in the community structure of the network.

There are various ways in which the community structure may change from time t to time $t+1$, also shown in the visualization:

Growth: a community can grow through new nodes.

Contraction: a community can contract through node removals.

Merging: two or more communities can merge into a single community.

Splitting: one community can split into two or more communities.

Birth: a new community can appear at a given time.

Death: a community can disappear at any time.

Resurgence: a community can only disappear for a period and come back later.

The problem of detecting communities in dynamic networks is more recent and there is not a single method yet that is considered generally accepted. Most approaches in the literature follow one of the following three general approaches.